

Ritesh Thawkar

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 Ritesh Thawkar |  Ritesh Thawkar |  Google Scholar |  Hugging Face

OBJECTIVE

Researcher focused on LLMs/VLMs, agentic RAG, and multimodal systems, with experience deploying real-world LLM applications and conducting research in self-evolving LLMs/VLMs and unified understanding-generation models. I am eager to contribute to collaborative teams pushing state-of-the-art multimodal AI.

EDUCATION

- **Mohamed Bin Zayed University of Artificial Intelligence** August 2024 - Present
Master of Science in Computer Vision Abu Dhabi, UAE
 - GPA: 3.55/4.00
- **Dr. Babasaheb Ambedkar Technological University** February 2021 - May 2024
Bachelor of Technology in Computer Science and Engineering Lonere, India
 - GPA: 8.51/10.00

SELECTED PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

- [C.1] Omkar Thawakar, Shravan Venkatraman, Ritesh Thawkar, Abdelrahman Shaker, Hisham Cholakkal, Rao Muhammad Anwer, Salman Khan, Fahad Khan. (2026). **EvoLMM: Self-Evolving Large Multimodal Models with Continuous Rewards**. In *CVPR 2026 Findings*.
- [S.1] Abdelrahman Shaker, Ahmed Heakl, Jaseel Muhammad, Ritesh Thawkar, Omkar Thawakar, Senmao Li, Hisham Cholakkal, Ian Reid, Eric P. Xing, Salman Khan, Fahad Shahbaz Khan. (2026). **Mobile-O: Unified Multimodal Understanding and Generation on Mobile Device**. *arXiv preprint*.
- [C.2] Wafa Al Ghallabi, Ritesh Thawkar, Sara Ghaboura, Ketan Pravin More, Omkar Thawakar, Hisham Cholakkal, Salman Khan, Rao Muhammad Anwer. (2025). **Fann or Flop: A Multigenre, Multiera Benchmark for Arabic Poetry Understanding in LLMs**. In *EMNLP 2025*.
- [C.3] Omkar Thawakar, Dinura Dissanayake, Ketan Pravin More, Ritesh Thawkar, et al. (2025). **LlamaV-o1: Rethinking Step-by-step Visual Reasoning in LLMs**. In *Findings of ACL 2025*.
- [C.4] Sara Ghaboura, Ketan Pravin More, Ritesh Thawkar, Wafa Al Ghallabi, Omkar Thawakar, Fahad Shahbaz Khan, Hisham Cholakkal, Salman Khan, Rao Muhammad Anwer. (2025). **Time Travel: A Comprehensive Benchmark to Evaluate LMMs on Historical and Cultural Artifacts**. In *Findings of ACL 2025*.
- [C.5] Omkar Thawakar, Dmitry Demidov, Ritesh Thawkar, Rao Muhammad Anwer, Mubarak Shah, Fahad Shahbaz Khan, Salman Khan. (2025). **Beyond Simple Edits: Composed Video Retrieval with Dense Modifications**. In *ICCV 2025*.

SELECTED PROJECTS

- **MBZUAI Web Agent Suite:** 
Tools: [Agentic RAG, Web Agents, LLM Orchestration, FastAPI, PostgreSQL]
 - Built an LLM-driven agentic pipeline that resolves MBZUAI website queries using retrieval, tool-use, and response verification.
 - Deployed the system in production, serving thousands of users and iterating prompts, tools, and routing based on real traffic.
 - Developed an interaction analytics dashboard to track user intents, content gaps, and unresolved queries for continuous Agentic RAG tuning.
- **Lawa Agent Studio:** 
Tools: [LLM, RAG, Workflow Orchestration, React, FastAPI, Vector DBs]

- Built a platform for creating customizable LLM-powered agentic chatbots with configurable tools, tone, and knowledge sources.
- Implemented ingestion and indexing pipelines for website and document sources to enable fast, accurate LLM responses.
- Added analytics and evaluation hooks to measure resolution rates, user satisfaction, and retrieval quality for LLM improvements.
- **Lawa.ai:** 
Tools: [LLM, Document Retrieval, Vector Databases, FastAPI, PostgreSQL]
 - Designed and developed LLM-powered web assistants to improve user interaction and engagement for websites.
 - Implemented a document retrieval system using vector databases to ground LLM responses and improve query resolution accuracy.
 - Developed backend workflows and API integration using FastAPI and websockets for scalable deployment.
- **Nutrigenics.care:** 
Tools: [LLM, Recommendation System, Django, Bootstrap, PostgreSQL]
 - Designed and developed the platform's web infrastructure to manage user and recipe data efficiently, enhancing scalability and usability.
 - Integrated NutritionGPT to power LLM-based recipe recommendations from image inputs and user preferences.
 - Implemented a personalized recommendation system leveraging LLMs to tailor nutrition and recipe suggestions to individual needs.
 - Optimized backend workflows using Django and PostgreSQL to ensure efficient data handling and faster response times.

EXPERIENCE

- **Mohamed Bin Zayed University of Artificial Intelligence** *September 2024 - Current*
Research Assistant *Abu Dhabi, UAE*
 - Worked on developing a Visual Reasoning Benchmark for testing the reasoning capabilities of VLLMs.
 - Assisted in the research on Composed Video Retrieval, where we proposed the idea of using dense video descriptions to retrieve the target video with desired compositional changes.
 - Worked on developing document retrieval systems and web agents for the lawa.ai AI platform.
 - Worked on the development of Nutrigenics.care AI startup platform.
- **Quantiphi Analytics**  *February 2024 - May 2024*
Framework Engineer Intern *Remote*
 - Worked on building data-ingestion pipelines to handle customer data for small to medium businesses.
 - Involved in developing ML-driven systems to draw insights from the company data.
 - Optimized data processing workflows to improve pipeline efficiency, reducing overall processing time by 20%.

SKILLS

- **Programming Languages:** Python, JavaScript, C
- **Web Technologies:** Django, Flask, FastAPI, React.js, HTML5, CSS3, jQuery, Bootstrap, TailwindCSS
- **Database Systems:** Sqlite3, MySQL, PostgreSQL
- **Data Science & Machine Learning:** Pytorch, Tensorflow, Numpy, Pandas, Matplotlib, Scikit-learn
- **Cloud Technologies:** AWS, Azure
- **DevOps & Version Control:** Git, Github

ADDITIONAL INFORMATION

Languages: English (Intermediate), Hindi (Advanced), Marathi (Advanced)

Interests: Reading, Music, Sports (Cricket, Badminton)